# STORE ANALYTICS

## Performance Assessment Metrics

A Practical Handout for Data Science Practitioners

### Topics Covered

(1) Holdout Validation
(2) K-Fold Cross-Validation
(3) ROC Curve
(4) AUC — Area Under the Curve
(5) Bias-Variance Tradeoff

# Introduction — The Store Analytics Problem

Imagine you are a data scientist at RetailCo, a national grocery chain with 300 stores. Management wants to predict which customers are likely to churn — that is, stop shopping at RetailCo in the next 90 days — so the marketing team can intervene with personalised offers.

You build a binary classification model (churn = Yes / No) using features such as:

- Average weekly spend over the last 6 months
- Number of store visits per month
- Basket diversity (number of unique product categories purchased)
- Distance from nearest store
- Recency of last purchase

> **Learning Objective**
> By the end of this handout you will understand five essential evaluation concepts — Holdout Validation, K-Fold Cross-Validation, ROC Curve, AUC, and the Bias-Variance Tradeoff — applied directly to this churn-prediction scenario.

## (1) Holdout Validation

### What is it?

Holdout validation splits your labelled dataset into two non-overlapping subsets: a training set and a test set. The model is trained exclusively on the training data and evaluated on the held-out test data — data it has never seen.

### Diagram

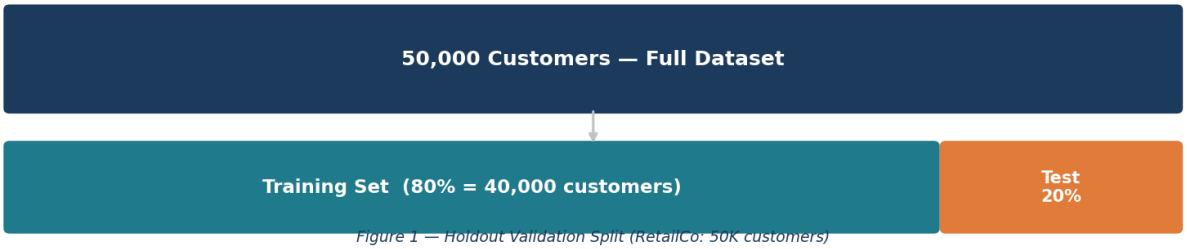



*Figure 1 — Holdout Validation Split (RetailCo: 50K customers)*

### RetailCo Example

You have transaction records for 50,000 customers over the past year. You split them:

| Split | Customers | Purpose |
|---|---|---|
| Training Set (80%) | 40,000 | Model learns churn patterns |

| Split | Customers | Purpose |
|---|---|---|
| Test Set (20%) | 10,000 | Simulate real-world performance |

You train a logistic regression on the 40K customers and then evaluate accuracy on the 10K test customers — this gives an honest estimate of how well the model will perform on new, unseen shoppers next quarter.

### Pros & Cons

| Advantages | Limitations |
|---|---|
| Fast and simple to implement | High variance — a lucky/unlucky split can skew results |
| Computationally cheap | Wastes 20% of labelled data for training |
| Easy to explain to stakeholders | Unreliable with small datasets |

**Rule of Thumb**

Use holdout when you have >= 10,000 samples and need a quick baseline. For RetailCo's 50K customers it is a safe starting point, but churn rates vary by season — a time-based split (e.g., train on Jan-Sep, test on Oct-Dec) is more realistic than a random split.

## (2) K-Fold Cross-Validation

### What is it?

K-Fold Cross-Validation divides the dataset into K equally sized folds. The model is trained K times; each time, one fold is held out as the validation set and the remaining K-1 folds form the training set. The final performance score is the average across all K runs.

### Diagram — 5-Fold Structure

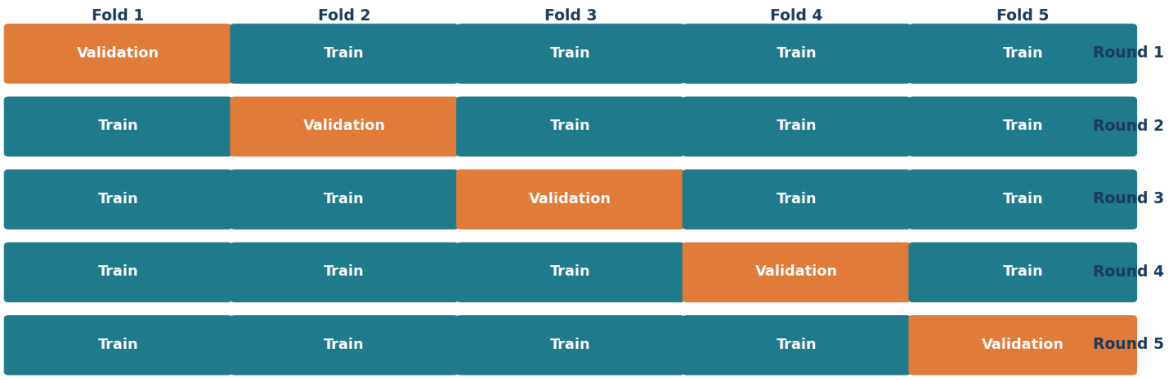



*Figure 2 — 5-Fold Cross-Validation: each fold serves as validation once*

### RetailCo Example — 5-Fold

With 50,000 customers split into 5 folds of 10,000 each, every customer is used for validation exactly once. Final AUC = average of AUC scores from all 5 rounds — a far more stable estimate than a single holdout split.

| Round | Training Folds | Validation Fold |
|---|---|---|
| Round 1 | Folds 2, 3, 4, 5 (40K) | Fold 1 (10K) |
| Round 2 | Folds 1, 3, 4, 5 (40K) | Fold 2 (10K) |
| Round 3 | Folds 1, 2, 4, 5 (40K) | Fold 3 (10K) |
| Round 4 | Folds 1, 2, 3, 5 (40K) | Fold 4 (10K) |
| Round 5 | Folds 1, 2, 3, 4 (40K) | Fold 5 (10K) |

### Why K = 5 or K = 10?

- K = 5 balances computation cost versus variance in the estimate.
- K = 10 gives lower bias (more training data per fold) but is 2x slower.
- K = N (Leave-One-Out) is ideal for tiny datasets but prohibitively slow for 50K records.

**Practical Note for Retail**

If your dataset has seasonal churn patterns (e.g., churners spike after the holiday season), use Stratified K-Fold, which ensures each fold contains the same proportion of churners (~15% in RetailCo's case). This prevents a fold from being overwhelmingly non-churners.

## (3) ROC Curve — Receiver Operating Characteristic

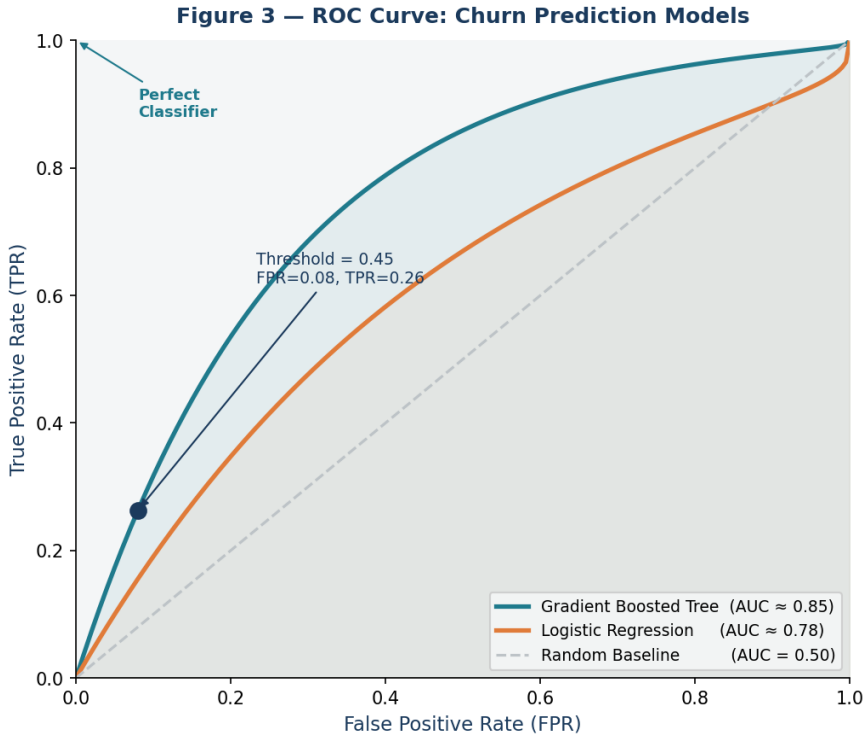
### What is it?

The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at every possible classification threshold. Unlike accuracy, it is threshold-independent and handles class imbalance well.

| Term | Formula | RetailCo Meaning |
|---|---|---|
| True Positive (TP) | — | Correctly predicted churner |
| False Positive (FP) | — | Loyal customer flagged as churner |
| False Negative (FN) | — | Churner we missed |
| True Negative (TN) | — | Correctly identified loyal customer |
| TPR / Recall | TP / (TP + FN) | % of churners correctly caught |

| Term | Formula | RetailCo Meaning |
|---|---|---|
| FPR / Fall-out | FP / (FP + TN) | % of loyal customers wrongly flagged |

## ROC Curve — RetailCo Churn Models




The curve bowing toward the top-left corner indicates better model performance. The diagonal dashed line represents random guessing (AUC = 0.50). The annotated point shows our chosen operating threshold at FPR = 0.08, TPR = 0.72 — balancing marketing spend vs. churn coverage.

**RetailCo Threshold Decision**

Marketing can afford to send a voucher to ~5,000 customers (10% of base). At threshold = 0.45, the model flags 5,200 customers with TPR = 72% and FPR = 8%. Lowering to 0.30 catches more churners (TPR = 85%) but flags 14% of loyal customers, wasting budget. The ROC curve makes this trade-off visual.

# (4) AUC — Area Under the ROC Curve

## What is it?

AUC summarises the entire ROC curve into a single number between 0 and 1. It equals the probability that the model ranks a randomly chosen churner higher than a randomly chosen loyal customer — a threshold-independent, intuitive metric.

## AUC Scale & Confusion Matrix

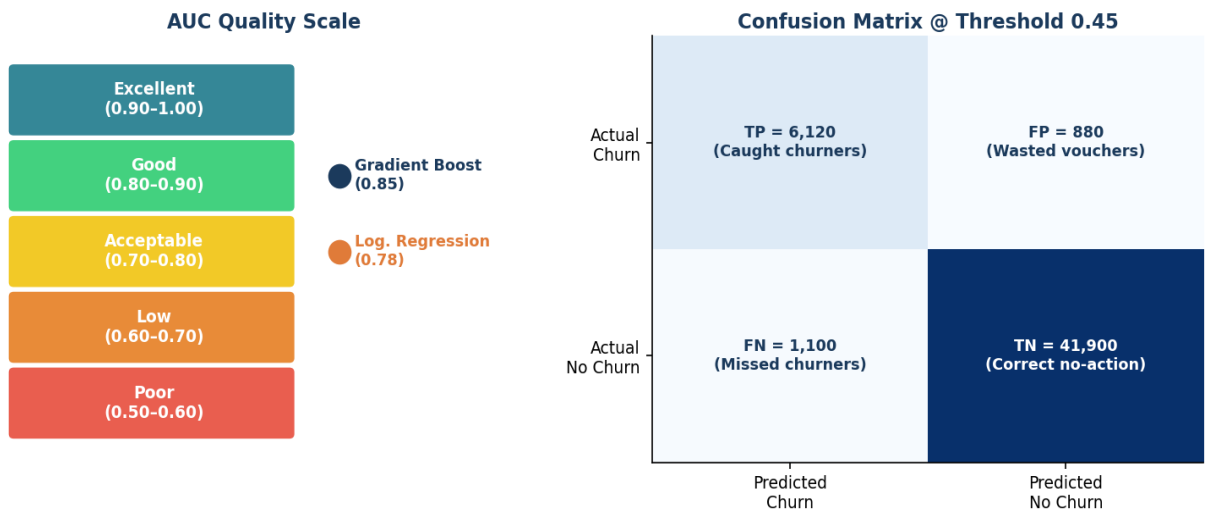



*Figure 4 — AUC Scale & Confusion Matrix (RetailCo, 50K customers)*

Left panel: The AUC quality scale shows where our two models sit. The Gradient Boosted Tree (AUC = 0.85) outperforms Logistic Regression (AUC = 0.78), despite both having ~88% raw accuracy.

Right panel: The confusion matrix at threshold = 0.45 shows 6,120 churners correctly caught, 880 wasted vouchers (FP), and 1,100 missed churners (FN). This drives the business cost calculation.

| AUC Score | Interpretation | RetailCo Context |
|---|---|---|
| 1.00 | Perfect model | Never happens in practice |
| 0.90 - 0.99 | Excellent | Strong signals in transaction data |
| 0.80 - 0.89 | Good | Solid production-ready model |
| 0.70 - 0.79 | Acceptable | Reasonable, room to improve |
| 0.60 - 0.69 | Poor | Limited practical value |
| 0.50 | Random baseline | Do not deploy |

**The Class Imbalance Trap**

RetailCo's churn rate is 15%. A naive model that predicts 'no churn' for every customer achieves 85% accuracy but catches zero churners (AUC = 0.50). Always use AUC alongside accuracy — or Precision-Recall AUC for very imbalanced data.

# (5) Bias-Variance Tradeoff

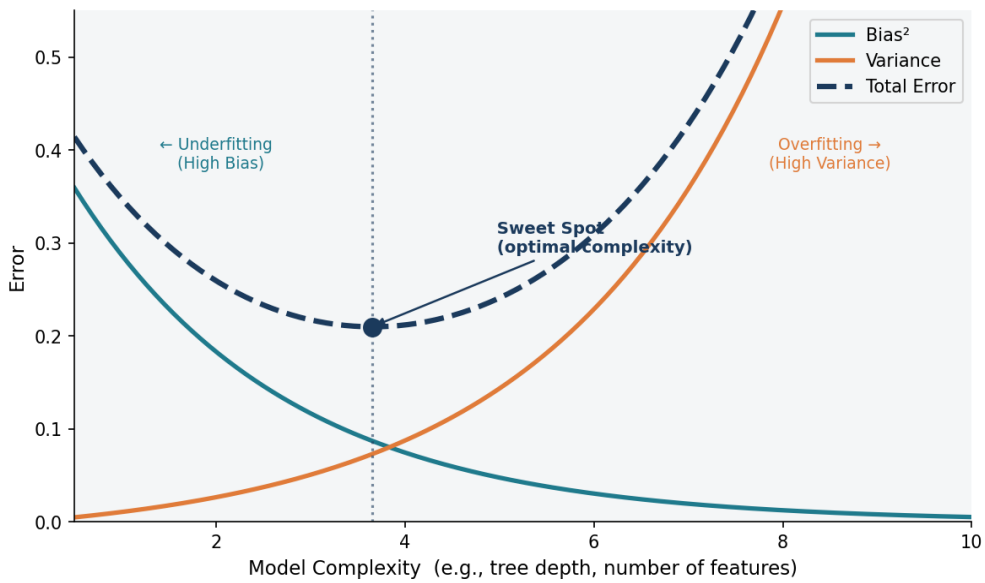
## The Core Concept

Every model's prediction error decomposes into three parts: Bias (error from wrong assumptions), Variance (sensitivity to training data fluctuations), and Irreducible Noise. The goal is to minimise their combined total.

| Component | Definition | Caused By |
|---|---|---|
| Bias | Error from wrong assumptions — model too simple | Underfitting |
| Variance | Sensitivity to training data fluctuations | Overfitting |
| Irreducible Noise | Random error inherent in the data | Cannot be reduced |

## The Tradeoff Curve

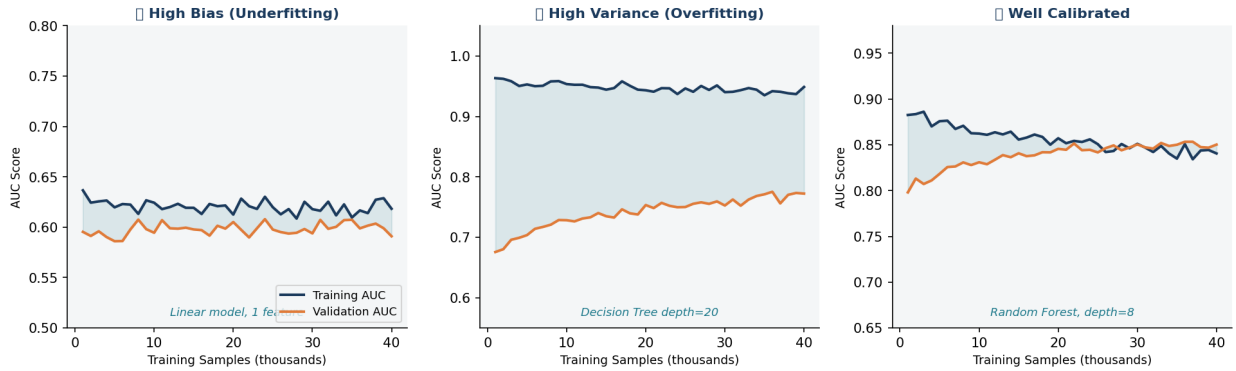
**Figure 6 — Bias-Variance Tradeoff Curve**




As model complexity increases (e.g., deeper trees, more features), Bias decreases but Variance rises. Total error is minimised at the 'sweet spot' — the optimal complexity for a given dataset and task.

## Diagnosing with Learning Curves

Figure 5 — Learning Curves: Diagnosing Bias & Variance in Churn Models




Plot training vs. validation AUC as training size grows. Three patterns emerge:

- High Bias: Both curves plateau low (~0.62 AUC). Adding more data barely helps. Fix: increase model complexity or engineer better features.
- High Variance: Training AUC is very high (~0.97) but validation AUC is much lower (~0.75) and the gap persists. Fix: add regularisation, prune the tree, or use ensemble methods.
- Well Calibrated: Both curves converge to a high AUC (~0.84-0.86). This is the production-ready state.

## RetailCo Model Comparison

| Underfitting | Overfitting |
|---|---|
| High Bias (Underfitting)<br>Linear model using only recency | High Variance (Overfitting)<br>Decision Tree with depth = 20 |
| Train AUC: 0.61 \| Test AUC: 0.60<br>Misses spend-visit interactions | Train AUC: 0.99 \| Test AUC: 0.67<br>Memorises individual quirks |
| Fix: add features, boost complexity | Fix: prune, regularise, ensemble |

**RetailCo Best Practice**

Use 5-Fold Cross-Validation to detect variance early. If the standard deviation of AUC across folds is > 0.03, the model is likely high-variance. A Random Forest with max_depth=8 and 300 estimators achieved AUC = 0.85 +/- 0.01 across folds — a well-balanced production result.

# Quick Reference Summary

| Concept | Key Question Answered | RetailCo Application |
|---|---|---|
| Holdout Validation | How does the model perform on unseen data? | Quick baseline on 10K held-out customers |
| K-Fold CV | Is my performance estimate stable? | 5-Fold CV to tune hyperparameters without waste |

| Concept | Key Question Answered | RetailCo Application |
|---|---|---|
| ROC Curve | What threshold best fits my business goal? | Balance marketing budget vs. recall of churners |
| AUC | How good is the model overall? | Compare models fairly on imbalanced churn data |
| Bias-Variance | Is my model too simple or too complex? | Learning curves to diagnose and fix misfits |

## When to Use What — Decision Guide

Use the following workflow when building a new model at RetailCo:

- Start with Holdout — get a fast, rough benchmark on a time-based 80/20 split.
- Switch to K-Fold (K=5) when comparing multiple models or tuning hyperparameters to avoid overfitting to the validation set.
- Plot the ROC Curve to determine the optimal classification threshold given the cost of false positives (wasted vouchers) vs. false negatives (missed churners).
- Report AUC as the primary metric to leadership — it is intuitive and threshold-independent.
- Monitor Bias-Variance by comparing training vs. cross-validation AUC; use regularisation, pruning, or ensembling to correct imbalances.